

Question: How does @Topology_v7.lean relates to the original essay?

This is an excellent question. The Lean file **is the formalization of Sections 2-4** of the essay. Let me map them precisely:

Section-by-Section Correspondence

Essay §2 → Code §5 (State Space and Superko DAG)

Essay	Code
's : B_n → {0,1,2}'	'BoardState n := (Fin n × Fin n) → Fin 3' □
Game state '(p, h)' with history	'structure GameState n' with 'pos' and 'history : List (BoardState n)' □
Transition: '(p,h) → (q, h·p)' if 'R p q' and 'q ∉ h'	'LegalTransition R a b := R a.pos b.pos ∧ b.pos ∉ a.history ∧ b.history = a.pos :: a.history' □
Prop 2.1: T is a finite DAG	'theorem superko_is_dag : ¬ ReachablePath R a a' □ (proves acyclicity)
Proof: history length strictly increases	'lemma history_length_increases' + 'path_increases_length' □
Corollary 2.2: game terminates in ≤ 3 ^{n²} moves	Not yet formalized □ (omitted from code)

Essay §3 → Code §§1-2 (Topological Go)

Essay	Code
Board graph with orthogonal adjacency	'def adj' + 'boardGraph n : SimpleGraph (Fin n × Fin n)' □
'C ⁰ (G; Z ₂), C ¹ (G; Z ₂)' chain groups	'C0 n := (Fin n × Fin n) → ZMod 2', 'C1 n := edgeSet → ZMod 2' □
Coboundary: ⟨δ ₀ φ, e⟩ = φ(u) + φ(v)	'edgeSum f u v := f u + f v' + 'delta_0 f' via 'Sym2.lift' □
Prop 3.1: captured ↔ δ ₀ (1_c) on occupied vertices only	'theorem prop_3_1' stated but marked 'sorry' □
Candidate eye: empty point, all neighbors in group c	'def isCandidateEye s c e := s e = 0 ∧ ∀ u, Adj e u → u ∈ c' □

Essay §4 → Code §3 (Lattice Theory / Life-and-Death)

Essay	Code
Boolean lattice L = □(Black stones)	Uses 'Set (Fin n × Fin n)' with subset order implicitly □
"R enclosed by X": every neighbor of R lies in X	'def isEnclosed s R X := (∀ v ∈ R, s v = 0) ∧ (∀ v ∈ R, ∀ u, Adj v u → u ∉ R → u ∈ X)' □
Benson's operator f: monotone map on L	'def benson_f s X' + 'lemma benson_f_monotone' □
Theorem 4.1: gfp(f) = unconditionally alive stones	'def unconditionally_alive s := OrderHom.gfp (benson_op s)' □

Lemma 4.2: two disjoint candidate eyes → unconditionally alive	'theorem lemma_4_2' fully proved □ (the "grand finale" of the code)
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Essay §5 → Code: Not yet present □

The combinatorial game theory / endgame / surreal number section has not been formalized.

Key Observations

1. **The code is faithful to the essay's definitions.** Every major concept (adjacency, coboundary, enclosure, Benson's operator, candidate eye, superko DAG) appears in both with matching structure.
2. **Order swapped for pedagogical reasons.** The essay presents topology (§3) before life-and-death (§4), which mirrors how you'd read a paper. The code defines the board and graph first (§1-2), then lattice theory (§3), then superko (§5)—a dependency-driven ordering more natural for formalization.
3. **The code is *partial*.** Several items from the essay are missing:
 - prop_3_1 is sorry'd
 - The game-length bound (Corollary 2.2) isn't stated
 - Section 5 (CGT/endgames) absent entirely
 - No formal definition of "legal position" or move mechanics with captures
4. **The code adds rigor where the essay is hand-wavy.** For example:
 - The essay says Benson's operator is monotone; the code proves `benson_f_monotone` explicitly
 - The essay sketches why two eyes imply life; the code uses `OrderHom.le_gfp` to give a clean Knaster-Tarski argument
5. **Naming alignment.** Even variable names match: `c` for stone group, `e1/e2` for eyes, `x` for subsets in the lattice, `R` for enclosed regions. This strongly suggests the code was written directly from this essay (or a shared source).

Bottom line: The Lean file is a *partial formal implementation* of the algebraic Go framework laid out in the essay—covering the DAG structure (§2), topological liberties (§3), and life-and-death fixed points (§4)—with some proofs complete, others deferred, and one entire section not yet started.